

Clave 10F02 "D"

$$1) \quad v = 65000 \text{ km/h} \rightarrow 18055.5 \text{ m/s}$$

$$t = 5 \text{ segundos}$$

$$x = v \cdot t$$

$$x = 18055.5 \frac{\text{m}}{\text{s}} \times 5 \text{ s}$$

$$\boxed{x = 90277.5 \text{ metros}}$$

$$2) \quad v_0 = 150 \text{ km/h} \rightarrow 41.6 \text{ m/s}$$

$$v_f = 0$$

$$t = 58 \text{ s}$$

$$a) \quad a = \frac{v_f - v_0}{t} \rightarrow a = \frac{0 - 41.6 \text{ m/s}}{58 \text{ s}} \rightarrow \boxed{a = -0.71 \text{ m/s}^2}$$

$$b) \quad v_f^2 = v_0^2 + 2a\Delta x \quad \Delta x = \frac{-v_0^2}{2a} \rightarrow \Delta x = \frac{-41.6^2 \text{ m/s}^2}{2 \cdot (-0.71 \text{ m/s}^2)}$$

$$\boxed{\Delta x = 1218.7 \text{ metros}}$$

$$3) \quad v_0 = 0$$

$$v_f = 27 \text{ km/h} \rightarrow 7.5 \text{ m/s.}$$

$$t = 5 \text{ segundos}$$

$$a) \quad a = \frac{v_f - v_0}{t} \rightarrow a = \frac{7.5 \text{ m/s} - 0}{5} \rightarrow \boxed{a = 1.5 \text{ m/s}^2}$$

$$b) \quad \Delta x = \frac{v_f^2}{2a} \rightarrow \Delta x = \frac{7.5^2}{2 \cdot (1.5)}$$

$$\boxed{\Delta x = 18.75 \text{ metros}}$$